AI-Native RAN: Operator's Perspective on 6G Day 1 Standardization

Towards AI-Native RAN — China Mobile, arXiv:2507.08403

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Table of contents

1 View at a glance **2**

2 Motivation: Limitations of add-ons and the significance of Day 1 **2**
2.1 Why it must be native . . . 2
2.2 Day 1 Why the Standard Is Special . . . 3

3 Three essential capabilities **3**
3.1 Capability 1 — AI-Driven RAN Processing / Optimization / Automation . . . 3
3.2 Capability 2 — Efficient, Controllable, and Reliable AI LCM . . . 4
3.3 Capability 3 — AIaaS Provisioning . . . 5

4 Day 1 Standardization Principles **5**

5 Proposed Architecture: AI Node + 6gNB **6**

6 Large-scale Field Trial **7**
6.1 Case 1 — Differentiated Service Guarantee . . . 7
6.2 Case 2 — Analysis of root cause of defect experience . . . 7
6.3 Case 3 — network energy saving . . . 8
6.4 Limitations revealed by the trial . . . 8

7 Research gap and thesis differentiation connection **8**
7.1 Positioning Summary . . . 9

8 Metadata **10**

1 View at a glance

This is a paper on **operator-perspective 6G AI-Native RAN standardization position** by authors from China Mobile (Nan Li, Qi Sun, Chih-Lin I, etc.). Rather than a study proposing new algorithms, it is closer to a blueprint outlining "how to natively incorporate AI into the first release (Day 1) standard of 6G" based on 2G to 5G operation and standardization experience.

The core arguments are two.

- In 5G, AI/ML was an **add-on** built on top of existing structures and was bound by legacy architecture, protocol, and flexibility constraints. 6G must natively embed AI from the start.
- At this time, **AI for RAN** (enhancing RAN performance with AI) and **RAN for AI** (RAN providing compute/connectivity for pervasive AI) Must be pursued simultaneously

As specific contributions, we present (1) a summary of the three essential capabilities of AI-Native RAN, (2) a proposal for a Day 1 architecture including a new RAN node called **AI Node**, and (3) verification of a large-scale field trial on the scale of 5,000+ 5G-A base stations.

> The values in this document were verified by comparing them with the original text (arXiv:2507.08403v1).

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2 Motivation: Limitations of add-ons and the significance of Day 1

2.1 Why it must be native

The paper views the problems faced by 6G RAN as areas where AI excels at solving. Representative examples include (i) low-complexity signal processing in massive MIMO, where computational complexity increases rapidly with hundreds of antenna elements, and (ii) large-scale, non-linear problems or those where mathematical modeling itself is difficult, such as radio resource allocation and scheduling in large-scale UEs and complex topologies . In these problems, real-time algorithms are forced to compromise on performance, but AI can bypass this through learning.

2.2 Day 1 Why the Standard Is Special

The core argument is that the first release of each generation serves as the standard for actual large-scale commercial deployment. While standards evolve at a cycle of approximately 1.5 to 2 years per release, operators deploy networks much less frequently due to TCO. In 4G/5G as well, the first releases (Rel-8, Rel-15) were the primary targets for deployment, and subsequent release features received little vendor support or operator purchase.

As an example, **High-Speed Rail (HST)** is cited. China's HST network is expected to operate at speeds of approximately 48,000 km and up to 350 km/h as of 2024, but severe Doppler shift compensation was only introduced after Rel-16, resulting in low utilization in actual live networks. This is a warning that if the same thing happens with 6G, the "AI-as-afterthought" limitations will solidify in 6G as well, without being able to implement AI on Day 1.

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3 Three essential capabilities

The paper defines AI-Native RAN as having three capabilities (Fig. 1).

3.1 Capability 1 — AI-Driven RAN Processing / Optimization / Automation

Apply AI across L1/L2/L3. Summarize the strengths of AI into three points: learning in domains where explicit modeling is impossible, real-time decision-making in large-scale, high-dimensional environments, and prediction and proactive adaptation based on complex spatiotemporal patterns.

Table I is the core, and it maps use cases along two axes.

- **AI for resources control & optimization (L2/L3)**: load balancing, mobility management, network energy saving, network slicing, capacity & coverage optimization, **multi-user scheduling & resource allocation**, QoS/QoE & SLA assurance.
- **AI for air interface (L1)**: positioning, beam management, CSI compression/prediction, AI receiver/transceiver, PA non-linearity compensation, interference prediction, RS overhead reduction.

For each use case, the inference location (BS/UE/LMF-side), model type and scale (DNN/LSTM/DRL/GNN/Transformer, `<1M~10M-1B`), and standardization status (3GPP Rel-18/19, O-RAN, *6G candidate use case*) are indicated together.

Key Point. `multi-user scheduling and resources allocation` is classified as a **6G candidate use case**, and The proposed model is also on a rough scale at the level of `DNN(<1M)`, `DRL(<10M)`. In other words, this paper itself is scheduling It merely lists "areas where AI can be applied," leaving blank which signal to use as a utility and how to solve it(→ Section 7).

3.2 Capability 2 — Efficient, Controllable, and Reliable AI LCM

It is the three pillars of data collection, model management, and distributed collaborative computing.

(a) Data collection. The 5G RAN protocol is designed for communication purposes and is unsuitable for AI data collection. The control plane is limited by payload size, and the user plane is disadvantageous for RAN AI because data is terminated at the core's UPF and is subject to charging; furthermore, mass collection risks depleting radio links and impeding basic connectivity. Three solutions are proposed:

- *Dedicated AI radio bearer*: A dedicated bearer dynamically configured and managed by the RAN to carry training data, models, performance metrics, etc. A new data tunnel for AI data exchange between RAN nodes is also included.
- *Cross-domain data collection*: RAN collects real-time traffic characteristics and service QoE of the application server. For example, optimize PDU set mapping and DRB configuration by receiving the **TTFT (time to first token)** of LLM inference.
- *Task-driven customizable collection*: Supports **10 ms or less** latency compared to the OAM method with an approximately 15-minute cycle, and Reduces unnecessary resource waste by collecting only on a specific UE unit with a filter condition (e.g., `UE velocity > 250 km/h`).

(b) Model management. Since off-line trained models experience performance degradation due to real-world distribution mismatches, online training, fine-tuning, and pre-validation are required. Performance monitoring should be viewed in three dimensions: **model performance / network performance / resource utilization** (Table II). This includes model context (ID/version) synchronization upon UE migration, performance-triggered/periodic retraining, RAN Digital Twin-based pre-deployment verification, and support for FL and large foundation models.

(c) Distributed collaborative computing. Solves the lack of compute, energy, and data in individual nodes through collaboration with a central AI node. Defines three modes: collaborative *training*, *inference*, and *fine-tuning*, and proposes resource pooling and dynamic scheduling in which gNB and UE register compute capabilities to form a logical compute pool.

3.3 Capability 3 — AIaaS Provisioning

The operator evolves from a connectivity provider → an edge AI compute provider → an edge AI service enabler. Three service modes (Fig. 8):

- **Connection assurance for AI services**: Extension of QoS framework for opaque flows such as multimodal input and model update, **Support for new traffic patterns** (lateral flow of embodied intelligence, *uplink-heavy · bursty* traffic of personal assistant), **deterministic latency** (assistant `< 200 ms`, embodied AI `< 100 ms`) guaranteed.
- **Data services**: Exposes internal metrics such as traffic load and link quality, and sensing data such as mobility and location. For example, a robotic system adjusts the sensing/ update cycle to avoid packet loss.
- **RAN AI computing services**: Exposing heterogeneous compute such as GPUs/NPUs to external edge AI (requires joint communication-computing scheduling).

All three of these are based on **open API framework** standardization.

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4 Day 1 Standardization Principles

1. **Day 1 Standard Range Determines Effectiveness** — Late-Introduced Features Have Low Usage (HST Case).
2. **Forward compatibility is essential** — (hardware) You must be able to run subsequent release software on the equipment where it was installed first, and Recover investment and idle compute per site through centralized resource management + distributed processing. (software/model) Since AI models can change their structure or be retrained, the interfaces between processing blocks and protocol layers must be clearly defined.
3. **Unlocking Commercial Value with RAN AI Service Exposure** — Since AI compute is expensive and consumes a lot of power, Expose idle AI capacity to external AI services like a cloud to create revenue streams beyond connectivity.
4. **3GPP-led + SDO collaboration** — 3GPP is responsible for end-to-end architecture, interfaces, and interoperability, and ITU is responsible for the umbrella framework/terminology (FG-AINN), while ETSI and O-RAN are responsible for RAN cloud and RAN application (API). Two options were presented: a proposal in which 3GPP defines the AI Node and its interface, and an alternative defined by O-RAN.

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5 Proposed Architecture: AI Node + 6gNB

The most specific contribution of the paper (Fig. 9). The key is **functional dispersion based on timeliness**.

Component	Location/Form	Function
RAN OAM	central cloud, non-real-time	global supervision · coordination, MDA, etc. proprietary AI
AI Node	locally centralized, sufficient AI compute (edge platform)	data collection, model management, **model repository**, **AI/ML training** (both L1/L2/L3), **L3 inference**, RAN capability exposure
6gNB	distributed, UE connection	Basic communication (dynamic scheduling) + **L1/L2 real-time inference** + corresponding LCM/data collection
6G UE	Terminal	Basic communication + (Limited) inference · Data collection

The AI Node and 6gNB are connected to a new RAN interface with **one-to-many cardinality**.

Design Logic:

- Centralize heavy **training to AI Nodes** → Minimize AI compute requirements for 6gNB and aggregate sporadic training Increase resource efficiency with **statistical multiplexing** (Enabling inference services on BSs without AI → Low-cost deployment).
- **L1/L2 inference requiring real-time is placed in 6gNB.** The original text states that this functional split is "Guarants real-time operation of L1/L2"
- Remaining compute is exposed to OTT/RAN Apps and converted into an AIaaS revenue source (exposed to the core via new SBI, and to OAM via the management interface).

  This **timeliness-based split** exactly aligns with the position that "L1/L2 real-time control must be node-local." This can be used as a standardization basis for the argument that instead of externalizing into a Near-RT RIC xApp, it should be collocated to a node (→ Section 7).

6 Large-scale Field Trial

Phase 3 trial conducted by China Mobile in 2024. Verification through the actual deployment of **CCU (Centralized Computing Unit)** corresponding to AI Nodes.

- **Phase 1** (2024.1~12, China **31 cities · 5000 5G-A base stations**): Centralized-Distributed AI Coordination + Verification of Three Use Cases.
- **Phase 2**: Expanded to 475,000 gNB, high-density environments such as stadiums and performance venues, AI-based RRM, experience-centric subscription model.
- **Phase 3**: Nationwide deployment, UAV mobility management, embodied intelligence QoE, RAN capability exposure(AIaaS).

6.1 Case 1 — Differentiated Service Guarantee

AI classifies traffic in real-time (1,000+ app types, accuracy exceeding 95%; 4,800 concurrent users in 8 gNB) and triggers QoS optimization when degradation is detected. The core formulation is **maximizing utility expectation**:

$$\hat{\mathbf{a}}_u \;=\; \arg\max_{\mathbf{a}\in\mathcal{A}} \; \mathbb{E}[\,U(\hat{\mathbf{q}}_u(\mathbf{a}, \mathbf{x}_u))]$$

- $\mathbf{a}$: action(scheduling weight, resource allocation) for user u
- $\mathbf{x}_u$: Observation context (application type, RAN metric, mobility state)
- $\hat{\mathbf{q}}_u(\cdot)$: AI-predicted service-level quality vector (latency, throughput, QoE)
- $U(\cdot)$: Utility function reflecting operator policy (maximize QoE or minimize latency)

decreased by **33.6% / 30.5%** in poor coverage (RSRP ≤ -105 dBm) and heavy load (PRB $\geq 50\%$) , with an average of **43.0 ms → 32.0 ms (25.6%)**. QR code latency decreased by an average of **18.5 ms → 14.5 ms (21.9%)**. As a side effect, as links stabilized, apps increased their resolution from 720p to 1080p/4K, resulting in a **3~6% increase in traffic** per scenario.

6.2 Case 2 — Analysis of root cause of defect experience

Fusion of multi-layered UE · network · application data to classify user/cause categories:

$$\widehat{y}_u \;=\; f_\theta\big(\mathbf{x}_u^{\text{App}},\, \mathbf{x}_u^{\text{RAN}},\, \mathbf{x}_u^{\text{UE}},\, \mathbf{x}_u^{\text{Mob}}\big)$$

XGBoost classifies user type (indoor/outdoor/high-speed) with **over 90% precision** using App layer (TCP RTT · packet loss), RAN (SINR · RSRP · handover statistics), UE attributes, and mobility (grid positioning, $50\,m \times 50\,m$) as input. Diagnostic accuracy **+20%** and classification recall **+30%** compared to rule-based.

6.3 Case 3 — network energy saving

Model total power consumption based on approximately 20 types of ES techniques in the time, frequency, space, and power domains:

$$\mathbb{E} = \sum_{n=0}^{N} \sum_{m=0}^{M} \sum_{t=0}^{T} \left[P^{PA}(P, m, t) + P^{\text{transceiver}}(m, t) + P^{\text{digital IF}}(m, t, C) \right] + \sum_{n=0}^{N} \sum_{t=0}^{T} \left[P^{\text{baseband}}(t, C) + P_0 \right]$$

Traditional ES relies on manual parameters such as fixed time windows and static thresholds, making it sluggish to dynamic traffic. Two AI solutions:

- *Solution 1*: Incorporate AI into NMS to predict traffic using 15-minute cell-level data and dynamically adjust ES thresholds → **26.6%** reduction.
- *Solution 2*: Place CCU in BS to recognize service types in real-time, extend packet aggregation and sleep for delay-tolerant traffic → **34.16%** reduction.

(+5.32% and +12.88% respectively compared to non-AI methods.)

6.4 Limitations revealed by the trial

Cross-platform collaborative inference was blocked due to the **lack of standard interfaces between CCU and multi-vendor gNB**, and openness and reusability were limited due to the **lack of modular API encapsulation** of AI functions. → This directly leads to the justification for Day 1 standardization (interoperable AI service framework).

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7 Research gap and thesis differentiation connection

This paper is a **high-level blueprint**, so it does not address "how L2 schedulers incorporate token importance or traffic prediction into their utility ." This is where the contribution of importance-aware MAC scheduling comes in.

What this paper provides (Anchor)	Thesis gap
Utility Expectation Maximization scheduling formulation for Case 1 ($\arg\max_{\mathbf{a}} \mathbb{E}[U(\hat{\mathbf{q}}_u(\mathbf{a}, \mathbf{x}_u))]$) — form validated by the operator	No design injects **token importance (model entropy · attention)** as a utility signal into the input $\hat{\mathbf{q}}_u$ of $U(\cdot)$ → Extended to a formulation that directly optimizes end-to-end task utility/resource
Multi-user scheduling specified as a **6G candidate use case** in Table I (DNN/DRL, coarse scale)	Specific utility mapping and evaluation metrics (task utility instead of BLER) for **importance-aware L2 scheduling** that aligns with PRB/TB granularity are TBD
Connection assurance: embodied intelligence lateral flow, UL-heavy · bursty traffic, deterministic latency(`<100 ms` embodied)	There is no scheduling policy that directly relieves offloading and UL pressure in multi-robot (stationary arm/AMR/drone) **heterogeneous mobility** → motivation can be adopted as is
Cross-domain data collection: LLM **TTFT** → PDU set mapping · DRB optimization	**predictive resource demand estimation** from token content (Stage 3) is not implemented
Task-driven 10 ms data collection, filter(e.g., velocity>250 km/h)	**proactive PRB pre-allocation(chance-constrained, Stage 4)** that incorporates predictions in advance is not implemented
Timeliness-based functional split: L1/L2 real-time inference is 6gNB (node-local)	Cited as the **standardization side's basis** for the claim that "sub-TTI scheduling must be GPU-collocated and cannot be externalized with Near-RT RIC xApps"

7.1 Positioning Summary

- **Value as a Citation Anchor**: China Mobile · Chih-Lin I's operator/standardization vision → Authoritative basis for the AI-RAN framework.
- **Motivation Match**: UL-heavy · bursty traffic, robot offloading, deterministic latency directly match the use case.
- **Clarity of the gap**: This paper only presents the form of "solving scheduling as a utility," and, We leave it blank to **what (token importance)** and **how to fill that utility through prediction and pre-allocation**. Stage 3 (predictive demand estimation) and Stage 4 (proactive pre-allocation) are those blanks.

8 Metadata

- Title: *Towards AI-Native RAN: An Operator's Perspective of 6G Day 1 Standardization*
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